Sibtain Raza

Lead Machine Learning Engineer

Solution-oriented ML engineer with 8+ years of experience across healthcare, law enforcement, digital marketing, and logistics. Have designed end-to-end ML workflows for audio, document image, text, and real-time video data — from medical dictation and clinical entity extraction to document information extraction and, most recently, LLM/RAG-based agents that let stakeholders query documents, images, video, and audio in natural language.

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WORK EXPERIENCE

Sr. AI Consultant TargetZero, *USA (Remote)*

August 2025 – Present

- Architected and deployed a multi-agent AI assistant for law-enforcement investigations using Pydantic AI, FastAPI, and AWS Bedrock (Claude 3.5/3.7 Sonnet), supporting multi-turn conversational querying over enterprise incident data.
- Built a real-time streaming architecture using Redis Streams and Server-Sent Events, supporting reliable, reconnectable agent sessions without duplicated inference cost.
- Designed context-management strategies to keep long, multi-turn agent conversations coherent while controlling token cost at scale.
- Implemented enterprise-grade security for a multi-tenant law-enforcement environment, including fail-closed access control and input guardrails against prompt injection.
- Built an internal evaluation framework combining automated regression checks with LLM-based quality grading to catch issues before release.
- Added observability tooling — audit logging, cost tracking, and execution tracing — to support compliance and debugging.
- Built RAG-based semantic search over incident narrative text using LanceDB for vector search and AWS Bedrock Titan embedding models.
- Built data connectors integrating CAD, Jail Management Systems, and Law Enforcement Management Systems, removing the need for manual data entry.

Lead ML Ops Engineer HighPlains Computing, *Los Angeles, CA (Remote)*

October 2023 – August 2025

- Built "Hercule," an LLM-powered analytics platform for law enforcement that ingests audio, video, images, and PDF documents and converts them into structured, queryable data:
  - Used self-hosted Whisper for speech-to-text and a Qwen vision-language model for document/image understanding to extract structured data from unstructured evidence.
  - Built a RAG system using ChromaDB for vector storage and AWS Bedrock models for embeddings, queryable through a LangGraph-based chatbot interface (Hercule).
  - Integrated AWS Textract for structured document extraction as part of the ingestion pipeline.
- Built "Lili," a healthcare solution that automatically extracts ICD-10 codes, CPT codes, medications, and anatomy from physician notes and automates entry of that data into an EHR system:
  - Built and maintained a scalable data pipeline for Lili using AWS services including AWS Batch, AWS SageMaker, AWS ECS, AWS Athena, and DynamoDB.
  - Worked extensively with FHIR APIs to create and manage encounters, appointments, and other operations for clinical notes processing.
- Delivered analytics for Careem (ride-hailing) as a client engagement, building reports on user login activity, platform usage, and engagement across the product's user base.
- Optimized real-time and batch processing workflows for scalability and cost efficiency.

Sr. Software Engineer (ML Ops) Point of IT, *Islamabad, Pakistan (Remote, US-based company)*

October 2021 – October 2023 (promoted from Software Engineer, ML Ops, Oct 2021 – Aug 2022)

- Designed and maintained CI/CD pipelines using Terraform (Infrastructure as Code) to provision and manage AWS environments.
- Built and optimized AWS Batch-based compute pipelines for large-scale model training and inference; containerized ML services with Docker for scalable batch-processing jobs.

Delivered DevOps and applied-ML engineering across three client engagements:

  - **MoodMD —** Maintained Terraform infrastructure pipelines and Python DevOps scripts to monitor and ensure reliable execution of production ML models.
  - **Boston University (BUCAS) —** Designed and deployed a CI/CD pipeline for the university's DevOps workflows.
  - **Gardiant —** Built Terraform infrastructure-as-code and Python automation for CI/CD pipelines deploying AWS Lambda functions and ECS applications.

AI/ML Engineer AI Bees, *Hamburg, Germany (Remote)*

September 2021 – August 2022

- Developed marketing automation tools using LangChain-based pipelines for prospecting and lead generation, dockerized and served via FastAPI.
- Built a response-classification model (PyTorch) via text augmentation and fine-tuning of DistilBERT, achieving a 0.91 F1-score on test data.
- Worked on named entity recognition to extract relevant information from text using a PyTorch-based XLM-RoBERTa model.
- Used self-hosted LLMs such as GPT-J and GPT-Neo to generate conditioned text for marketing campaigns and cold emails.

Data Scientist PackageX, *Islamabad, Pakistan*

March 2021 – September 2021

- Built parcel-tracking extraction for a mailroom OCR service, extracting sender/receiver details from parcel images across multiple languages using OCR and LayoutLMv3.
- Deployed models with reduced inference latency using ONNX format and PyTorch Serve, exposed via FastAPI; built TensorFlow Lite computer-vision models for mobile deployment.

Data Scientist VisionX, *Islamabad, Pakistan*

January 2020 – March 2021

- Built OCR and document image-analysis pipelines for structured/unstructured data extraction, achieving a 0.92 F1-score at field level across thousands of image templates.
  - **Invoice Automation —** Automated extraction of line items, totals, and vendor details from invoices and receipts using OCR combined with LayoutLMv2 and LayoutLMv3.
  - **Insurance Cards & Driving Licenses —** Extracted structured fields from insurance cards and driving licenses using Tesseract OCR combined with a PICK (graph-based) model.
- Built a PDF redaction pipeline to detect and mask PII across financial and healthcare documents using OCR and layout-based detection models, supporting GDPR compliance.
- Built the cloud architecture for a Voice Verification REST API using a GE2E-loss-based speaker embedding model, Docker, and Flask.

Jr. Data Scientist Care Cloud, *Islamabad, Pakistan*

June 2018 – January 2020

- Built an EHR virtual assistant chatbot using Rasa Core and Rasa NLU for intent detection and automated IVR call flows.
- Trained a multi-class classification model for imbalanced healthcare text data, using BERT to generate text embeddings and a custom fine-tuning layer, served via a dockerized Flask application on a Linux server.
- Worked on a speech recognition model using the PyTorch implementation of Deep Speech; annotated IVR call audio data and applied a statistical language model on top of Deep Speech to convert audio to text, then analyzed the text with spaCy and a custom classifier to extract relevant information.

- Developed REST APIs using Flask to create microservices responsible for fetching and updating data via SQLAlchemy.

SKILLS

Python	TensorFlow	PyTorch	Machine Learning
Pydantic AI	LangChain	LangGraph	Multi-Agent Orchestration
AWS Bedrock	RAG / Agentic AI	LLM-as-a-Judge	Eval Harnesses
Guardrails & Prompt-Injection Defense	RBAC & Multi-Tenancy	Observability	AsyncIO
FastAPI	Redis Streams	Server-Sent Events (SSE)	PyTorch Serve
LanceDB	Milvus	Chroma	NLP
Computer Vision	OCR	LayoutLM	Docker
Flask	Terraform	CI/CD	AWS
GCP	MySQL	PostgreSQL	DynamoDB
Redis	Elasticsearch	Rasa	

EDUCATION

BS Computer Science

FAST – National University of Computer and Emerging Sciences, Islamabad

2014 – 2018

LANGUAGES

Urdu (Native), English (Fluent)